

AI-Based Cybersecurity Threat Detection Web Application

Abstract

CyberShield is an AI-Based Cybersecurity Threat Detection Web Application developed using Python Flask and machine learning. The system accepts 15 network traffic features and predicts whether the given traffic is Normal Traffic or Cyber Attack Detected. The application also stores user-wise prediction history with date, time, country, and IP address. It provides chart analysis, CSV export, PDF report generation, malware scanning, chatbot support, profile page, and admin-controlled user access.

Problem Statement

Modern computer networks handle large volumes of traffic. Manual identification of suspicious packet flow, abnormal bytes per second, DDoS-like bursts, malicious file uploads, and unauthorized users is difficult. A web-based system is required to collect network features, classify threat behavior, store logs, and generate clear reports for analysis.

Objectives

- To classify network traffic as Normal Traffic or Cyber Attack Detected using selected traffic features.
- To provide a clean step-by-step flow: Login, Home, Prediction, Result, History, and Reports.
- To store prediction data user-wise with date, time, country, and IP address.
- To show visual chart analysis and provide CSV/PDF download options.
- To include a basic malware scanner for uploaded files.
- To include a chatbot that explains cybersecurity concepts and project features.
- To allow admin users to allow or block normal user access.

System Modules

- Login and Registration: authenticates users using username/email and hashed password.
- Home: introduces the project and provides access to main modules.
- Prediction: collects 15 network traffic values and sends them for classification.
- Result: displays final result, severity, confidence, score, recommendation, IP address, and country.
- History: displays all user prediction records with chart, PDF download, CSV export, delete, and clear options.
- Malware Scanner: checks uploaded file extension, size, hash, and suspicious keywords.
- Chatbot: answers questions about CyberShield modules and cybersecurity concepts.
- Report: explains the project clearly for academic submission and provides PDF download.
- Profile: shows logged-in user account information.
- Admin Panel: allows admin to view users and decide whether each user can access the application.

Technologies Used

Category	Tools / Purpose
Frontend	HTML, CSS, JavaScript, Chart.js
Backend	Python Flask
Database	SQLite
Machine Learning	scikit-learn Random Forest, joblib model file
Security	Password hashing, Flask session, role-based admin access
Reporting	FPDF for PDF generation and CSV export
File Analysis	Werkzeug secure upload, SHA-256 hash, static file checks

Architecture and Workflow

The application follows a simple web architecture. The user interacts with HTML templates. Flask handles routing and backend logic. SQLite stores users, detections, and malware scan records. The trained machine learning model predicts the traffic class. Chart.js displays visual analytics, while FPDF and CSV modules generate downloadable reports. Workflow: Login -> Home -> Prediction -> Result -> History -> Chart/PDF/CSV. Additional modules are Malware Scanner, Chatbot, Report, Profile, and Admin Panel.

Machine Learning Logic

The project uses a Random Forest based classifier saved as cyber_model.pkl. The prediction page sends 15 numeric traffic features to the model. Random Forest combines multiple decision trees and is suitable for classification because it handles multiple numeric features and gives stable output. If suspicious network patterns are detected, the result is shown as Cyber Attack Detected; otherwise it is shown as Normal Traffic.

Input Features

The 15 input features are Protocol, Flow Duration, Total Forward Packets, Total Backward Packets, Forward Packets Length, Backward Packets Length, Forward Packet Length Mean, Backward Packet Length Mean, Forward Packets/sec, Backward Packets/sec, Forward IAT Mean, Backward IAT Mean, Flow IAT Mean, Flow Packets/sec, and Flow Bytes/sec.

Normal vs Attack Explanation

Normal traffic usually has balanced packet flow, moderate packet rate, reasonable bytes/sec, and stable flow duration. Attack traffic may show abnormal packet bursts, very high bytes/sec, unusual packet imbalance, or suspicious flow timing. The final result is decided using the combined pattern of all 15 features, not only one input value.

Malware Scanner Explanation

The malware scanner is a basic static analysis module. It checks the uploaded file name, extension, file size, SHA-256 hash, risky extensions such as .exe, .bat, .js, .ps1, and suspicious keywords such as powershell or cmd.exe. Based on these checks, it generates a risk score and result. It is useful for academic demonstration but does not replace a professional antivirus engine.

Chatbot Explanation

The chatbot acts as a CyberShield help assistant. It can answer questions about project overview, flow, AI, Random Forest, dataset, 15 input features, prediction, normal traffic, cyber attack, result, history, chart, PDF, CSV, malware scanner, project report, profile, admin allow/block access, advantages, limitations, and future scope.

Admin Access Control

The admin panel is used for user management. Admin can view registered users and change their access status. If a user is blocked, that user cannot login or use the application. This gives the project a role-based access control feature.

Current User Statistics

Total Detections: 0

Normal Traffic: 0

Cyber Attacks: 0

Malware Scans: 0

Advantages

- Complete end-to-end project flow suitable for demonstration.
- User-wise history storage with date, time, IP address, and country.
- Chart-based analysis, CSV export, and PDF report download.
- Admin can allow or block users.
- Includes additional cybersecurity modules such as malware scanner and chatbot.
- Clean UI with animation and theme toggle support.

Limitations

- Prediction accuracy depends on the trained dataset and model quality.

- Malware scanner uses basic static analysis only.
- Real-time packet capture is not fully integrated in this version.

Future Enhancements

- Real-time packet capture and live traffic monitoring.
- Email OTP or password reset system.
- Advanced malware sandbox analysis.
- Deep learning based threat classification.
- Cloud deployment and real-time alert notification.

Conclusion

CyberShield demonstrates how machine learning, Flask web development, database storage, visualization, report generation, malware checking, chatbot assistance, and admin access control can be integrated into a complete cybersecurity academic project. It is suitable for project review because it has a clear workflow, practical modules, and understandable outputs.